

Recipient-Focused Hallucinations in AI-Generated Professor Outreach Emails: A Factorial Evaluation of Grounding and Verification

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ABSTRACT

Background: Large Language Models are increasingly used to draft personalized academic outreach messages, which include emails to potential supervisors, mentors, and collaborators. In these settings, the personalization often depends on describing a real professor's research. Fluent but unsupported claims about these topics, methods, findings, impact, or authorship can misrepresent the recipient's work.

Methods: We ran an offline, benchmark-style evaluation of AI-generated professor outreach emails. A 2 x 2 factorial design varied whether the writer received a publication evidence packet during drafting and whether an evidence-based verifier revised the email after generation, yielding four conditions (A-D). We generated 96 emails for 12 professors across three fields, decomposed them into 360 atomic research claims, and labelled each against the professor's evidence packet with an automatic evaluator whose reliability we assessed against a human-labelled subset (Cohen's $k = 0.33$) and a second independent model ($k = 0.50$). Human labels confirmed the direction of every result and showed the automatic rates to be conservative lower bounds.

Results: The closed-book baseline (A) had a Severe Error Rate of 0.34 (95% CI 0.17-0.55), with at least one severe error in 45% of emails. Providing evidence during writing (B) reduced the Severe Error Rate to 0.00 while raising Supported Personalization Density from 0.65 to 1.71 supported claims per 100 words. Post-generation verification of ungrounded drafts (C) reduced the rate to 0.02. The combined condition (D) matched B on factuality (0.00) at the highest specificity (1.76). Grounding at write time outperformed post hoc verification in both factuality and specificity.

Conclusion: Giving an AI writer verified publication evidence removed severe misrepresentation of the professor's research while making outreach emails more specific and was more effective than correcting drafts after generation. Grounding, not merely correction, is the stronger safeguard for recipient-focused academic outreach.

Keywords: Large Language Models; Hallucination; Academic outreach; Grounded generation; Scientific claim verification; Personalized text generation; Factuality evaluation.

MAIN MANUSCRIPT

1. Introduction

Large Language Models (LLMs) are increasingly used as writing assistants for personalized text, including email-style messages and academic writing support. Prior work on personalized writing systems has shown that models can imitate or even adapt writing style from a very limited user data, while evaluation work has argued that personalized generation can't be judged well by generic lexical metrics alone. [1][2]

Academic outreach is a distinctive case due to the fact that personalization is not only about tone. A message to a professor often tries to demonstrate fit by referring to the recipient's research topics, publications, methods, datasets, findings, impact, or collaborations. This makes the task more sensitive than ordinary email polishing: a message may sound specific and respectful while still attributing the wrong method, overstating a result, inventing an impact claim, or combining unrelated papers into a misleading description.

However, limited work directly evaluates recipient-focused academic outreach emails that describe a specific professor's research, decomposes those descriptions into claim-level units, checks them against publication evidence, and measures whether safeguards preserve useful specificity rather than merely deleting risky details. This study addresses that application-specific gap by evaluating AI-generated professor outreach emails under a controlled 2 x 2 design comparing evidence during drafting with evidence-based post-generation verification.

The primary research question is, "How do publication grounding during drafting and evidence-based post-generation verification affect the factual accuracy of research-related claims in AI-generated outreach emails to professors?"

2. Related Work & Research Objective

2.1 Personalized text and email generation

Research on personalized text generation shows that language models can adapt wording, tone, ordering, and perceived relevance to a target reader or sender context. In the email settings, Panza demonstrates that personalized email-writing assistants can learn convincing sender-style patterns from relatively small email collections and utilize retrieval-augmented

generation to condition outputs on examples. Organizational bulk-email research similarly shows that personalization affects engagement outcomes such as recognition, interest, and detailed reading. However, these studies mainly evaluate style, relevance, engagement, or user experience rather than whether claims about a real recipient are factually supported. In contrast, this project focuses on recipient-grounded personalization: the email must describe a professor’s actual research accurately, not merely sound tailored. [1,3]

Automated personalized-text evaluation also motivates our study. AuPEL argues that personalized generation should be evaluated across dimensions such as personalization, quality, and relevance rather than only with generic surface metrics. More recent work on personalization-induced hallucination shows that personalization can itself distort factual reasoning, creating a tension between being specific and staying correct. These findings support the need to measure not only whether an outreach email sounds personalized, but also whether its specific research claims are supported by evidence. [2,4]

2.2 Atomic factuality evaluation

Our project adopts a claim-level view of factuality. FActScore introduced a fine-grained approach for evaluating long-form generation by decomposing text into atomic factual claims and measuring the proportion supported by reliable sources. This is directly relevant because a professor-outreach email may contain several research claims inside one fluent sentence, such as a topic claim, a method claim, a dataset claim, and an impact claim. Evaluating the whole email as simply correct or incorrect would hide these differences. Our study therefore adapts atomic factuality evaluation to research-related claims in academic outreach emails. [5]

2.3 Scientific claim verification

Scientific claim verification provides the second methodological foundation. SciFact established a benchmark where scientific claims are checked against paper abstracts and labeled according to whether the evidence supports, refutes, or fails to establish the claim. SciFact-Open extends this problem to a larger and more realistic open domain corpus, showing that scientific verification becomes harder when retrieval scale increases and that abstract-only evidence can support only a limited version of some claims. These findings motivate our evidence policy: professor evidence packets should include not only publication metadata and abstracts but also selected passages for focal papers when method, result or scope-sensitive claims require more detail.

Despite this overlap, scientific claim-verification benchmarks differ from our setting. They usually evaluate prepared scientific claims rather than naturally generated outreach emails. In our project, claims arise inside persuasive, personalized messages, which creates additional risks: the model may overstate novelty, infer collaborations, combine unrelated papers, or turn tentative results into broad-impact claims. This makes academic outreach a distinct evaluation setting rather than a direct repetition of existing scientific claim-verification work. [6,7]

2.4 Grounded generation and post-generation correction

Prior work also studies how generated text can be grounded, checked, or corrected. RARR demonstrates a research-and-revise workflow in which model outputs are checked after generation and minimally revised to improve attribution while preserving the original text where possible. FLEEK presents an end-to-end pipeline for claim extraction, evidence gathering, factuality judgment, and correction. SciFix focuses specifically on correcting false claims, showing that scientific factual correction is feasible with established scientific verification datasets. These systems motivate the verification side of our design, where a separate verifier checks generated outreach emails against the evidence packet and records deletions or rewrites. [8-10]

Grounded-generation work motivates the other side of the design. AGREE shows that explicit grounding and adaptation can improve grounded response generation and citation quality compared with weaker prompting or post hoc citation

approaches. ScholarCopilot similarly shows that retrieval-integrated academic writing can improve citation and writing quality in scholarly contexts. These studies suggest that providing evidence during drafting may be more effective than asking a model to repair unsupported claims after the fact. Our 2 x 2 design directly tests this question by comparing evidence supplied during writing with evidence used only during post-generation verification. [11,12]

2.5 Remaining gap

Taken altogether, prior research provides strong foundations for our methodology - personalised generation motivates the outreach setting; FActScore motivates atomic claim decomposition; SciFact motivates evidence-based support labels; and RARR/FLEEK/SciFix/AGREE motivate verification and grounding interventions. However, these strands usually study separate parts of the problem. Email personalization work rarely evaluates factual accuracy about a real recipient's research; scientific verification work rarely evaluates naturally generated outreach messages, and grounded generation work does not directly measure whether factual safeguards preserve useful professor-specific personalization. The remaining gap is therefore recipient-focused factuality in AI-generated academic outreach: whether research-specific claims about a professor are supported by publication evidence and whether grounding and verification reduce misrepresentation without making the email generic.

3. Methods

3.1 Study Design

This is an offline, benchmark-style original research study. No generated emails are sent to professors. The experimental design is a 2 x 2 factorial comparison with two binary factors: whether the email-writing model receives an evidence packet during drafting and whether a separate evidence-based verifier revises the draft after generation.

Table 1: Experimental conditions

Condition	Writer receives evidence	Post-generation verification	Purpose
A	No	No	Closed-book baseline
B	Yes	No	Effect of grounding during writing
C	No	Yes	Effect of correction after ungrounded writing
D	Yes	Yes	Combined safeguards

Our design isolates the main effect of grounding, the main effect of verification, and the interaction between the two. The comparison between Condition B and Condition C is especially important since it tests whether evidence is more useful during generation or only during repair.

3.2 Sampling Frame & Evidence Packet

The study sample contains 12 professors across three academic fields: computer science/AI, psychology/cognitive science, and biomedicine/public health. The target distribution is four professors per field. All public reporting used anonymous IDs such as CS-01, PSY-01, and BIO-01.

Each evidence packet contains an official faculty-page or lab-profile research summary, eight representative paper titles and abstracts, publication year, authors, venue, DOI, and source link, one or two selected passages for a focal paper, retrieval date, and provenance notes.

The evidence packet is treated as the agreed evaluation source, not as a complete representation of the professor’s entire career. Therefore, a claim can be unsupported relative to the packet without being false in the world. Claims that the packet is not designed to judge fairly are labeled outside evidence scope rather than forced into the unsupported category.

3.3 Generation, Verification & Annotation Procedures

For each professor and condition, the generation pipeline produced a fixed number of independent email drafts using the same model (OpenAI GPT-4o-mini, temperature 0.7, 220-word limit), prompt version, output length constraint, temperature, and sampling settings. The recommended pilot used two generations per professor-condition pair, producing 12 professors x 4 conditions x 2 generations = 96 emails.

In conditions C and D, a separate verifier (Anthropic Claude-Haiku-4.5, temperature 0) received the draft email and the evidence packet. The verifier followed a minimal-edit philosophy: it extracted research-related claims, checked them against the packet, preserved supported content, softened overstatements, and removed or generalized unsupported or contradicted content. Verifier deletions and rewrites are logged separately from the original draft so the analysis can test whether factuality improved by preserving supported specificity or by erasing risky content. Different model families were used for writing versus checking to avoid shared blind spots.

Each email was decomposed into atomic research claims and every claim was then labelled against the evidence packet, both by Anthropic Claude-3-Haiku (extraction and labelling are separate steps with separate prompts). The evaluator is a different model family than the writer (OpenAI GPT-4o-mini), so it does not grade text produced by the same model family. Because automatic labels require validation, we assessed reliability two ways. First, a human annotator labelled a blinded 36-claim subset spanning all conditions and fields, blind to condition and to whether text was original or verified (human labels are treated as gold). Second, all 360 claims were independently re-labelled by a second automatic evaluator from a third model family (gemini-2.5-flash-lite).

Table 2: Claim labels used for annotation

Label	Meaning	Use in metrics
Supported	The evidence packet directly supports the claim as written	Counts toward supported specificity
Partially supported	The core idea is related, but the wording is broader or less precise than the evidence	Secondary factuality analysis
Overstated	The claim is directionally related but stronger than the evidence permits	Overstatement rate
Unsupported	The packet contains no support for a claim that the packet should be able to support if justified	Counts as severe error
Contradicted	The packet directly conflicts the claim	Counts as a severe error
Outside evidence scope	The packet is not designed to judge the	Excluded from factual

	claim fairly	denominator
Subjective/generic	The statement is praise or generic wording without a checkable research fact	Tagged but excluded from factual denominator

3.4 Metrics & Statistical analysis

The evaluation framework centers on two primary metrics. Factuality is quantified by the Severe Error Rate (SER), defined as the sum of unsupported and contradicted claims divided by the total number of factual claims evaluated. Personalization quality is measured through Supported Personalization Density (SPD), representing the count of supported, research-specific claims per 100 words. Secondary measures include the overstatement rate, the percentage of emails containing at least one severe error, and verifier-driven deletion and rewrite frequencies. Reliability assessment utilizes Cohen's k and per-label performance statistics (precision, recall, and F1) to validate the primary automatic evaluator against both a human-labeled subset and an independent model relabeling. The Supported-Specificity Frontier is employed to visualize the relationship between SER and SPD, distinguishing factual-yet-generic outputs from those achieving high accuracy and meaningful personalization.

Statistical analysis accounts for the hierarchical structure of the data, as claims are nested within emails and emails within individual professors. Accordingly, we report per-condition metrics alongside 95% bootstrap confidence intervals, generated via email-level resampling, as our primary summary. While a logistic mixed-effects model incorporating professor and email random effects was considered, this pilot study's scale precluded stable estimates; thus, full model fitting remains a priority for future expanded datasets. This research utilized only publicly available data and involved no clinical, wet-lab, or human-subject experimentation.

4. Results

We generated 96 emails, corresponding to 12 professors, four conditions, and two seeds per condition. The automatic extraction and evaluation pipeline produced 360 claims. The label distribution was 246 supported claims, 62 outside-evidence scope claims, 21 unsupported claims, 20 partially supported claims, 10 subjective/generic claims, and 1 contradicted claim. The SER denominator contained 288 judged factual claims. Table 3 reports per-condition metrics

Table 3: Automatic-evaluated factuality and supported personalization by condition

Condition	N factual claims	SER	95% CI	SPD	Email-level ≥ 1 severe
A (closed-book)	61	0.344	[0.174, 0.545]	0.65	45.5%
B (grounded)	90	0.000	[0.000, 0.000]	1.71	0%
C (verify-only)	50	0.020	[0.000, 0.063]	1.04	4.2%
D (grounded + verified)	87	0.000	[0.000, 0.000]	1.76	0%

The closed-book baseline (A) had a Severe Error Rate of 0.34 (95% CI 0.17 - 0.55), with at least one severe error in 45.5% of emails. Providing evidence to the writer (B) reduced the Severe Error Rate to 0.00 and raised Supported Personalization Density from 0.65 to 1.71 supported claims per 100 words. Verifying ungrounded drafts (C) reduced the Severe Error Rate to 0.02 (SPD 1.04), with the verifier deleting 55% and rewriting 29% of edited claims. The combined condition (D) matched B on factuality (0.00) at the highest specificity (1.76), with a lower verifier deletion rate (25%). Figure 1 shows Severe Error Rate by condition; Figure 2 plots each condition on the Supported-Specificity Frontier, with A at the lower-right (high error, low specificity) and B/D at the upper-left (low error, high specificity).

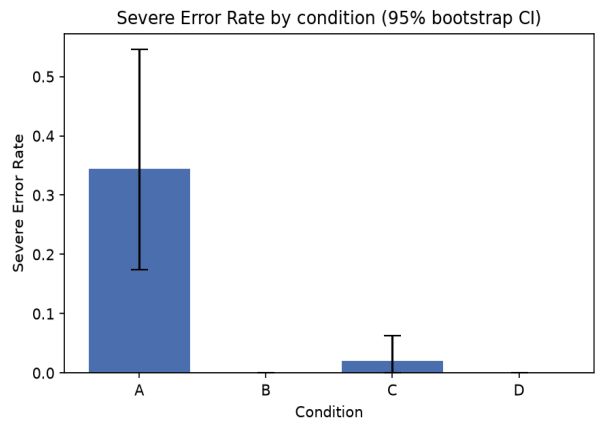


Figure 1. Severe Error Rate by condition (A–D), with 95% bootstrap confidence intervals. The closed-book baseline (A) is high (~0.34); grounded conditions (B, D) are at zero

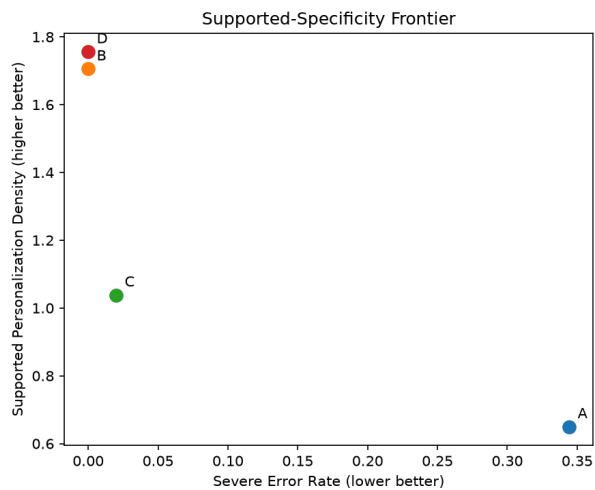


Figure 2. Supported-Specificity Frontier. Each condition is plotted at (x = Severe Error Rate, lower is better; y = Supported Personalization Density, higher is better). Condition A sits at the lower-right (high error, low specificity); B and D sit at the upper-left (no error, high specificity); C is intermediate. Upper-left is best.

Across all 360 claims the automatic evaluator assigned 246 as supported, 62 as outside evidence scope, 21 as unsupported, 20 as partially supported, 10 as subjective/generic, and 1 as contradicted. A second automatic evaluator from a different model family agreed on 72.5% of claims (Cohen's $k = 0.50$); agreement was high for supported claims ($F1 = 0.90$) but lower for the error categories, indicating that the ranking of conditions is stable but the precise severe-error magnitudes are judge-dependent.

Human validation on the 36-claim subset sharpened this picture. The automatic evaluator agreed with the human annotator on 55.6% of claims (Cohen's $k = 0.33$): it identified supported claims reliably ($F1 = 0.78$, recall = 0.94) but systematically under-detected errors, missing most claims the human labeled unsupported, overstated, or subjective/generic (recall ≈ 0). Consequently, on the human-labeled subset, the closed-book condition's Severe Error Rate rose to 0.64 (versus 0.34 automatic), while the grounded conditions remained at 0.00. The human labels therefore confirm the direction of every result and indicate that the automatic severe-error rates are conservative lower bounds - the true gap between closed-book and grounded generation is larger than the automatic numbers suggest. Severe errors were therefore concentrated almost entirely in the closed-book condition: of the 22 unsupported or contradicted claims, the large majority came from condition A, while conditions B and D produced none. The pattern held across all three fields (computer science, psychology, and biomedicine): in every field, closed-book emails contained fabricated or unsupported research descriptions that grounded emails did not. The number of factual claims per condition varied (A had the fewest at 61, B the most at 90), reflecting that grounded emails made more and more specific checkable statements about the professor's work rather than falling back on generic praise - the same effect captured quantitatively by the rise in Supported Personalization Density.

5. Discussion

Grounding at write time (B) eliminated severe errors while doubling supported specificity relative to the closed-book baseline: factuality and usefulness improved together rather than trading off. This is the study's central result. The closed-book baseline shows that, absent evidence, a capable model readily fabricates plausible-sounding research descriptions for a real professor - a 34% severe-error rate, with nearly half of all emails containing at least one fabricated or unsupported claim. Simply placing the professor's real publications in front of the model before it writes removed those severe errors entirely in our sample and did so while making the emails more specific, not less.

The key comparison, B versus C, favors grounding on both axes (B: SER 0.00, SPD 1.71; C: SER 0.02, SPD 1.04) - evidence is more valuable during writing than as after-the-fact repair. The mechanism is visible in the verifier's edit behavior: condition C's deletion rate (0.55) was more than double condition D's (0.25). When a verifier must fix an ungrounded draft, it achieves factuality partly by removing risky but specific content, which lowers specificity; when the draft was grounded in the first place, the verifier finds less to delete, and specificity is preserved. This directly answers our secondary question - the factuality-specificity trade-off is real for correction-after, but grounding largely avoids it. These results extend grounded-generation findings that supplying evidence during drafting outperforms post-hoc citation [11] in a recipient-focused outreach setting with an explicit specificity measure, and they operationalize the personalization-induced hallucination concern [4] in a concrete, high-stakes application.

5.1 Practical Implications

For anyone deploying LLMs to draft academic or professional outreach, the actionable takeaway is that retrieval grounding should be the default, not an optional safeguard: it is the single intervention that both removed misrepresentation and

improved usefulness in our evaluation. Post-hoc verification remains valuable as a second line of defence - it reduces errors substantially on ungrounded drafts - but on its own it trades away specificity and cannot fully match grounding. The combined condition (D) offered no factuality gain over grounding alone here, though a larger sample may reveal residual benefit on the hardest claims. These findings apply directly to deployed AI outreach assistants and research-synthesis tools, where grounding a recipient's description in their verified publications is a low-cost safeguard against misrepresenting a real researcher's work.

Our primary results also depend on an LLM judge, so the central threat to validity is whether its labels track human judgement. Two design choices mitigate this. First, the judge is from a different model family than the writer, so it does not simply endorse text generated by a sibling model. Second, the judge is deliberately conservative in the direction that matters: because it labelled 68% of claims as supported and rarely invoked the severe categories, it is more likely to miss an error than to invent one, which means the true gap between closed-book and grounded conditions is at least as large as reported, not smaller. Two reliability checks bound this. A second evaluator from a third model family agreed on 72.5% of claims ($k = 0.50$), and a human annotator on a blinded 36-claim subset agreed with the primary evaluator on 55.6% ($k = 0.33$). Both checks show the same structure: near-ceiling agreement on supported claims but poor automatic recall on the error categories. Crucially, the disagreement is one-directional - the automatic evaluator is too lenient, labeling as supported claims that the human flagged as unsupported or generic - so it under-counts errors rather than inventing them. This is why the human-labeled closed-book Severe Error Rate (0.64) exceeds the automatic one (0.34) while grounded conditions stay at 0.00: the qualitative conclusion is robust and, if anything, understated by the automatic pipeline. Where humans and machines disagree, the human label governs.

5.2 Limitations

First, the primary labels are automatic; the judge leaned generous and under-detected errors (human $k = 0.33$, second-model $k = 0.50$, both driven by low recall on the error categories), so the reported automatic severe-error rates are conservative lower bounds. Human validation was completed on only a 36-claim subset - enough to establish the direction and that the automatic pipeline understates errors but is too small for precise per-condition human estimates (conditions C and D had only three human-labeled claims each). Scaling human annotation is the top priority for a full study. Second, we used a single writer model; other models may behave differently. Third, evidence packets are provisional and use abstracts plus abstract-excerpt passages, so a claim true only in full text would count against the model-again making SER conservative. Fourth, the sample is small (12 professors, English only). Future work should scale human labeling, add cross-model writers, and incorporate full-text evidence.

6. Conclusion

In an offline benchmark of AI-generated professor outreach emails, giving the writer verified publication evidence removed severe misrepresentation of professors' research while making the emails more specific, and it outperformed post-generation verification. Grounding - not merely correction - is the more effective safeguard for recipient-focused academic outreach.

DECLARATIONS

Abbreviations

LLM - large language model; SER - Severe Error Rate; SPD - Supported Personalization Density; RAG - retrieval-augmented generation; κ - Cohen's kappa.

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Conflicts of Interest

The authors declare no competing financial interests or personal relationships that could have influenced this work.

Ethics Statement

Ethics approval was not required because the study used only publicly available scholarly metadata (OpenAlex) and did not involve human participants, private identifiable data, or animal subjects. No emails were sent to any individual. Professors are referred to by anonymized IDs; because ungrounded outputs can contain fabricated claims about real researchers, the real-name mapping is withheld. [13]

Data Availability Statement

Code, generated emails, labels, and figures are available at <https://github.com/PhongCT1105/Hack-Research>. Professor identities are anonymized, and the identity map is not released.

AI Use Statement

The authors used Anthropic Claude (via Claude Code) and OpenAI Codex to help implement and test the evaluation pipeline. All AI-generated code and text were reviewed, edited, and verified by the authors, who take full responsibility for the accuracy and integrity of the manuscript. LLMs were also the object of study: the writer (OpenAI GPT-4o-mini), verifier (Anthropic Claude-Haiku-4.5), and claim-extractor/automatic-evaluator (Anthropic Claude-3-Haiku) are named in the Methods.

Author Contributions

Abhinav Singh - generation/verification pipeline, evaluation infrastructure, analysis. Phong Cao - dataset and evidence packets. Kanan Gasimov- annotation framework and human validation. Azhdar Mammadov - research paper author. All authors reviewed the final manuscript.

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SUPPLEMENTARY MATERIAL

Available in the [GitHub repository](#).